

Essential Insights: Week 6



सिद्धिमूलं प्रबन्धनम्
भा. प्र. सं. इन्दौर
IIM INDORE

Why Agentic AI, Why Now?

The Evolution of AI Systems

- Classical Machine Learning focused on prediction
- Generative AI introduced content creation capabilities
- Agentic AI adds autonomous execution and workflow management

What Makes Agentic AI Different?

- Breaks goals into structured tasks through planning
- Uses external tools, APIs, databases, and enterprise systems
- Executes workflows instead of only generating outputs

Key Shift

- Move from AI-assisted work to AI-executed work

AI Agents vs Agentic AI

Understanding the Core Distinction

- AI agents perform individual tasks autonomously
- Agentic AI coordinates multiple specialised agents to achieve larger goals

AI Agent Capabilities

- Reasoning and planning
- Tool usage and memory
- Autonomous task execution

Agentic AI Capabilities

- Orchestration across multiple agents
- Workflow management and validation
- End-to-end process execution

Key Insight

- AI agents automate tasks; Agentic AI transforms workflows

Types of AI Agents

Agent Taxonomy

- Reflex Agents – React using predefined rules
- Model-Based Reflex Agents – Maintain environmental awareness
- Goal-Based Agents – Optimise actions to achieve objectives
- Utility-Based Agents – Balance multiple objectives and trade-offs

Enterprise Relevance

- Most organisations begin with goal-based agents
- Utility-based agents support complex optimisation problems

Business Value

- Increasing autonomy and decision sophistication across workflows

How AI Agents Work

The Agent Execution Loop

- Observe → Reason → Plan → Act → Reflect

Six Core Components

- Perception – Understands data, documents, and inputs
- Planning – Breaks goals into structured steps
- Reasoning – Determines actions and adapts dynamically
- Tool Calling – Interacts with external systems
- Memory – Maintains context and historical knowledge
- Learning – Improves performance through feedback

Outcome

- Transforms AI from a response engine into an execution system

Reasoning, Tool Use and Memory

Reasoning Approaches

- ReAct (Reason + Act) – Iterative thinking and action cycles
- ReWOO (Reasoning Without Observation) – Plans workflows upfront before execution

Tool Calling

- Connects agents to APIs, databases, CRM, ERP, and external systems
- Enables real-world actions rather than recommendations

Memory and Learning

- Short-term memory maintains interaction context
- Long-term memory stores organisational knowledge
- Learning creates continuous improvement over time

Business Impact

- Agents become smarter and more effective with experience

Agentic Architecture and Enterprise Design

A New Development Approach

- Traditional: Data → Model → Product
- Agentic: Business Outcome → Agent → Data → Model

Hierarchical Architecture

- Orchestrator agents manage workflows
- Executor agents perform actions
- Validator agents ensure quality and compliance

Key Advantage

- Adaptive, event-driven workflows that respond to changing conditions

Governance Need

- Human oversight becomes critical in high-risk processes

Multi-Agent Systems

Why Multiple Agents?

- Complex enterprise workflows exceed the capability of a single agent
- Specialised agents collaborate to achieve broader objectives

Core Agent Roles

- Orchestrator Agents – Planning and delegation
- Executor Agents – Task execution
- Validator Agents – Accuracy and compliance checks
- Communication Agents – User and system interactions

Enterprise Benefit

- Improved reliability, scalability, and workflow coordination

Agentic RAG: Smarter Retrieval Systems

Traditional RAG

- Retrieve information once and generate a response

Agentic RAG

- Dynamically decides what information to retrieve
- Selects sources and retrieval sequence adaptively
- Continuously refines retrieval strategy

Advanced Capabilities

- Adaptive planning
- Multi-source retrieval
- Corrective retrieval and self-correction

Business Value

- Supports complex enterprise queries spanning multiple knowledge repositories

Enterprise Applications and Governance

Workflow Transformation Across Functions

- Finance
- HR
- Procurement
- Supply Chain
- Customer Service

Key Outcome

- Shift from task automation to workflow compression and orchestration

Four Governance Principles

- Accountability
- Observability
- Human-in-the-loop oversight
- Data integrity

Critical Insight

- Scalable Agentic AI requires both capability and governance

Final Takeaways and Leadership Perspective

The Agentic AI Evolution

- Prediction → Generation → Execution
- AI increasingly performs work rather than supporting it

Core Building Blocks

- Perception, reasoning, planning, tool use, memory, and learning
- Multi-agent orchestration and Agentic RAG

Organisational Impact

- Workflow redesign rather than isolated automation
- Human–AI collaboration at enterprise scale

Leadership Imperative

- Focus on responsible design, governance, and integration of agentic systems into business processes

Key Question

- Not whether Agentic AI will be adopted, but how effectively it will be implemented and governed within the organisation